

Erdős Problem #1075

EP-1075

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $r \geq 3$. There exists $c_r > r^{-r}$ such that, for any $\epsilon > 0$, if n is sufficiently large, the following holds. Any r -uniform hypergraph on n vertices with at least $(1 + \epsilon)(n/r)^r$ many edges contains a subgraph on m vertices with at least $c_r m^r$ edges, where $m = m(n) \rightarrow \infty$ as $n \rightarrow \infty$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>